NFL BIG DATA BOWL 2023

Trench Matchup Win Probability

What sequence data reveals about pass-block outcomes that summary statistics miss

Kyle Smith

2021 SEASON · WEEKS 1–8

THE QUESTION

Does a model that watches the whole blocking rep predict its outcome better than one that only sees snapshots of it?

SOURCE	NFL Big Data Bowl 2023 player tracking data
SEASON	2021, Weeks 1–8 — the only season this tracking release covers
OUTCOME	A block is scored a loss on a sack, hurry, hit, or beaten-by-defender charge

MODEL A

XGBoost

Three photographs of the rep — distance, closing speed, and orientation at the snap, ~1 second in, and the final tracked frame.



gradient-boosted trees on engineered features

MODEL B

LSTM

The full film — every tracked frame from snap to outcome, read in order, roughly ten times per second.



a sequence model on the raw trajectory

WHAT EACH MODEL SEES

The same underlying variables, sampled two different ways

GEOMETRY	Distance, closing speed, orientation difference, and weight differential between blocker and rusher
PLAYER & MATCHUP	Blocker and rusher position, whether the block was flagged as assisted, and play-action
SITUATION	Down, yards to go, offense formation, and rusher count

XGBOOST SEES THIS

at three sampled instants: the snap, ~1 second in, and the final frame

LSTM SEES THIS

continuously, at every tracked frame from snap to outcome

THE RESULT

XGBOOST

.757

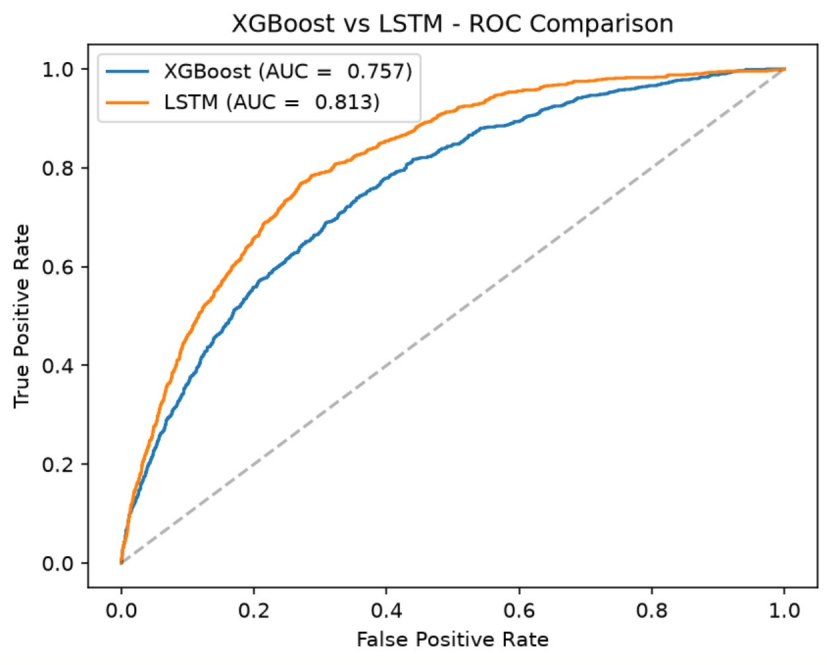
→

LSTM

.813

AUC on the same held-out, identifier-matched test set.

Recall on true blocking losses rose alongside it, roughly 56% → 67%.



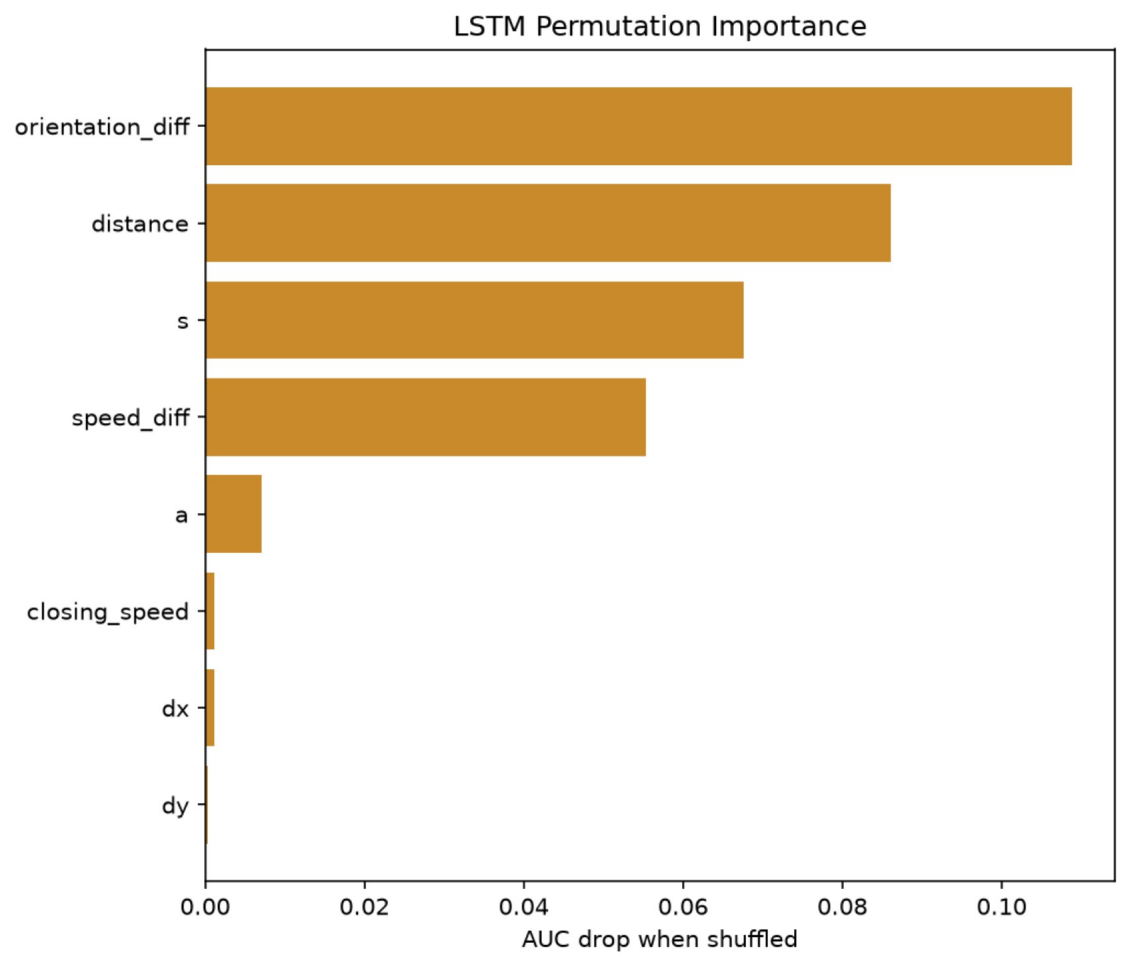
READING THIS CHART

The LSTM's curve sits above XGBoost's across nearly the entire range, not just at one threshold, meaning it ranks blocking outcomes better regardless of how conservative the flagging threshold is set.

AUC ran 0.806–0.821 across LSTM training seeds; this figure shows one representative run.

INTERPRETATION

What the LSTM actually learned



01 Both models agree: **orientation** — the blocker remaining square to the rusher — is the strongest signal of a win or loss.

02 They diverge on motion: XGBoost leans on a two-point closing-speed estimate; the LSTM reads raw speed continuously and barely needs it.

03 Direction alone (dx, dy) carries almost no weight — magnitude of separation and orientation is what matters, not which way it points.

MODEL TO PRODUCT

Three numbers, one dashboard

WIN RATE

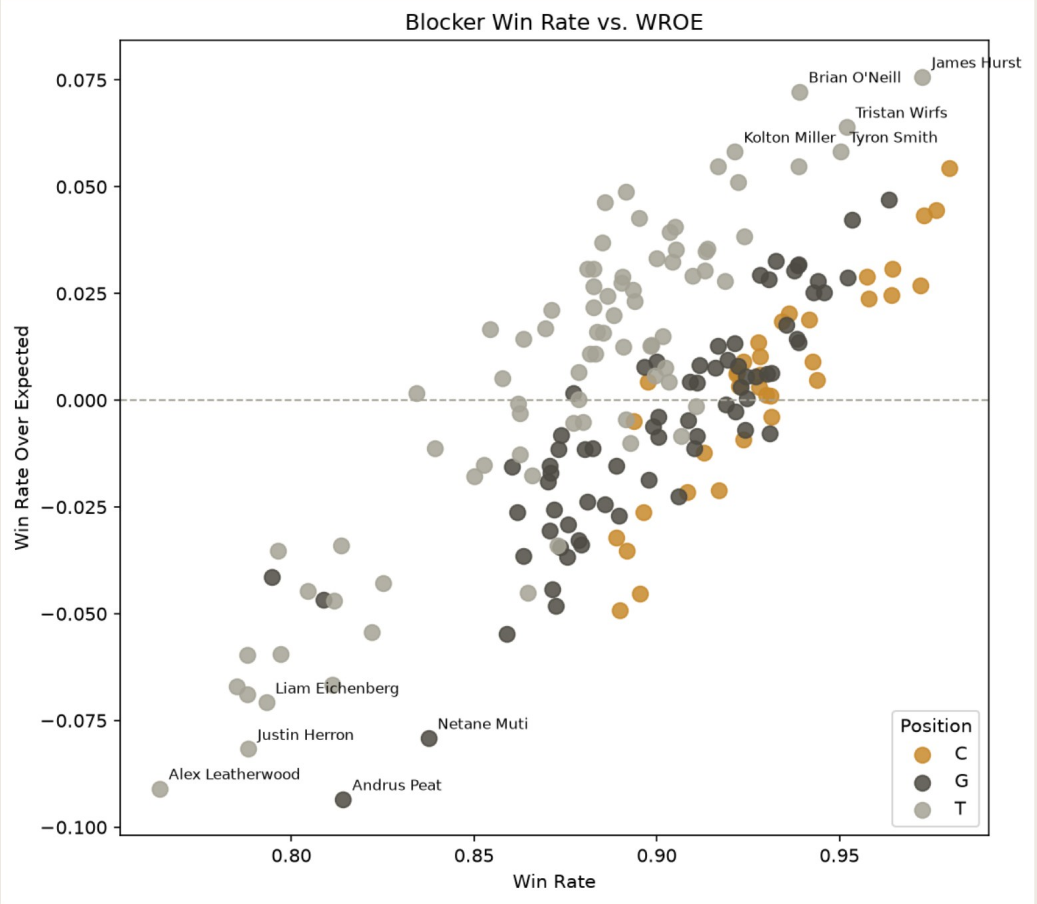
A blocker's actual share of clean reps this season

EXPECTED WIN RATE

What the LSTM predicted, given the difficulty of who he actually faced

WROE

Win rate over expected — actual minus expected, isolating skill from matchup difficulty



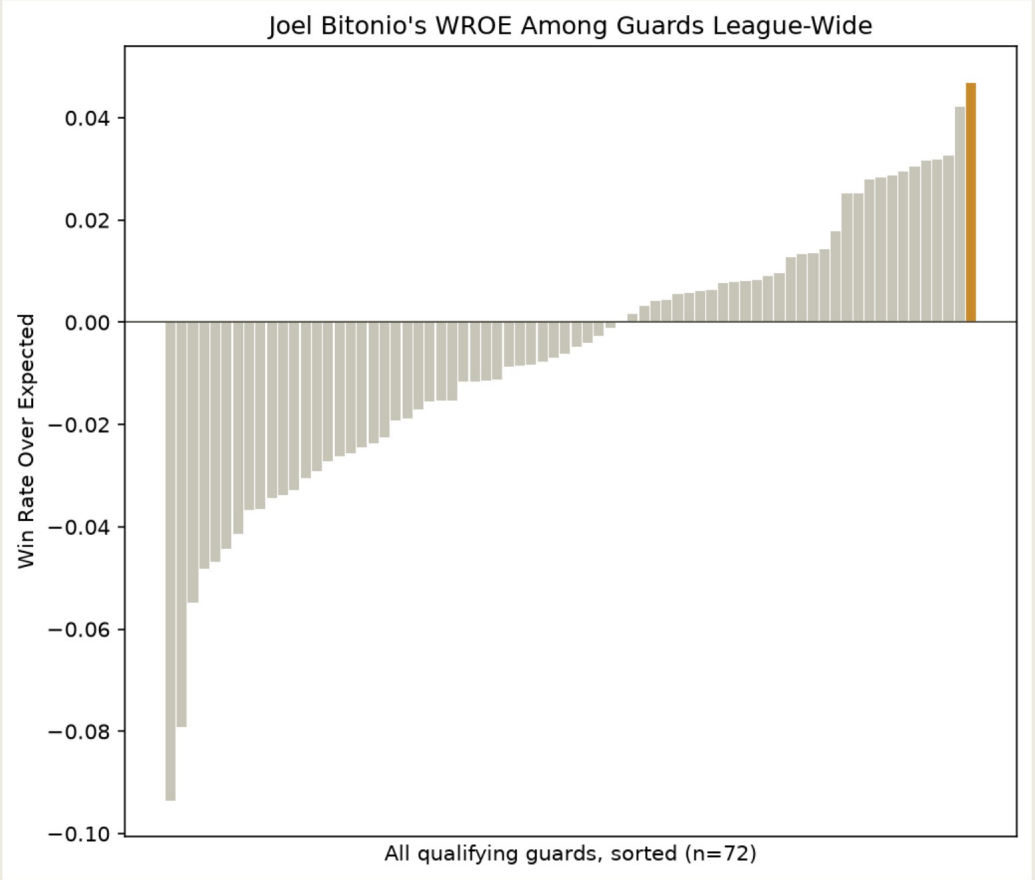
Tackles show the lowest raw win rate and the widest spread. Evidence the model inferred, without being told position, that tackles face tougher one-on-one matchups than guards or centers. Expected win rate absorbs nearly that entire gap, leaving WROE close to flat across positions.

IN PRACTICE

Joel Bitonio

WIN RATE	EXPECTED	WROE
96.3%	91.7%	+4.7%

Guards average slightly below expectation across the league — Bitonio posts the single highest WROE of any qualifying guard, 1st of 72.



Most guards cluster within a few points of zero. A long negative tail reaches nearly -9%, while a short positive tail tops out at Bitonio — standing alone, well clear of the next-closest guard.

TAKEAWAY

Pass-block outcomes are shaped by the trajectory of a rep, not just how it resolves — and a sequence model captures that a snapshot model structurally cannot.

.757 → .813	56% → 67%	42,080
AUC, SNAPSHOT VS. SEQUENCE MODEL	RECALL ON TRUE BLOCKING LOSSES	MATCHUPS SCORED ACROSS THE FULL SEASON